

LEKHANA P

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LeetCode | LinkedIn | GitHub | Portfolio

EDUCATION

JSS ACADEMY OF TECHNICAL EDUCATION
ELECTRONICS AND COMMUNICATION B.E
CGPA: 8.3

Bengaluru, Karnataka
2023 - 2027

JAWAHAR NAVODAYA VIDYALAYA
PCMB (Class 12)-CBSE
Percentage: 89.6%

Tumakuru, Karnataka
2022 - 2023

JAWAHAR NAVODAYA VIDYALAYA
(Class 10)CBSE
Percentage: 93.8%

Tumakuru, Karnataka
2020 - 2021

SKILLS

Programming Languages: Java, C++, Python, html
Libraries/Frameworks: Pandas, Scikit-learn, Matplotlib, TensorFlow, PyTorch
Tools / Platforms: Git & GitHub, VS CODE, Jupyter Notebook, Google Colab
Databases: SQL, MongoDB
Technical Concepts: DSA, OOPS, DBMS, REST APIs
AI / GenAI: LangChain, RAG (Retrieval-Augmented Generation), Google Generative AI, Vector Embeddings

PROJECTS

Legal Document Analyzer | Generative AI

November 2025-December 2025

FastAPI, LangChain, Google Generative AI, ChromaDB

- Built a scalable backend service using FastAPI to process and analyze legal documents.
- Designed a retrieval-augmented pipeline with optimized document search and response generation.
- Implemented modular APIs enabling structured querying and summarization of large text datasets. Reduced manual document analysis effort by automating risk detection and question and answer workflows.

Smart Specs | Assistive Wearable Device

August 2025-September 2025

Python, OpenCV, YOLOv8, BotSort, PyTorch, ESP32-CAM

- Developed a real-time monocular vision system for obstacle detection and spatial awareness using the pinhole camera model with focal length calibration and pixel-width-based depth approximation.
- Implemented multi-object tracking (BotSort) with a priority-based alert mechanism to maintain persistent object identities and announce only the nearest dynamic obstacle for efficient decision-making.
- Designed a modular, edge-ready architecture with scene-understanding triggers for contextual image description and future integration with embedded hardware (ESP32-CAM / edge AI devices).

Tour Buddy | Travel Assistance App

June 2025-July 2025

Next.js, Node Js, Google Generative AI, MongoDB

- Built a full-stack web application supporting dynamic itinerary generation.
- Designed backend services to handle asynchronous data fetching and API integration.
- Implemented database schema for efficient storage and retrieval of user travel data. Improved application responsiveness using optimized state management and API handling

CERTIFICATIONS

- Oracle Cloud Infrastructure AI Foundations - **ORACLE**

HONORS & AWARDS

- Awarded the prestigious **Reliance Undergraduate Scholarship**, for meritorious students across India for undergraduate studies.